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PROJECT AND TEAM INFORMATION

Project Title

Market Lens – Business Dashboard for Tracking Marketing Trends on Social Media Platforms

Student / Team Information

Team Name: **Team ID:**	Market Lens AJP-III-2026-T053
Team Member 1 (Team Lead)	*Gupta, Suhani – 2520230432* *guptasuhani117@gmail.com* 
Team Member 2	*Goswami, Angel – 2520230407* *angelgoswami864@gmail.com* 
Team Member 3	*Sharma, Urvi – 2520230239* *urvisharma070@gmail.com* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

Market Lens is a comprehensive, data-driven and trend monitoring platform which helps the user to identify emerging trends, analyze popularity, discover, monitor and understand the ongoing mania on different platforms all at once (on the same platform) with the help of social media data. The main goal of this project is to collect large amounts of social media data and present it in a simple and useful way through a single dashboard.

This dashboard gathers data anomalies such as popularity, categories, trend names, engagement metrics and other relevant information. The data is cached and stored methodically in a MySQL database, while the Spring Boot backend manages the business logic, filtering, REST APIs to establish a connection between database and frontend. The frontend is designed using React which provides an interactive UI which is simple and user-friendly where the client is able to explore, visualize and at ease in using the dashboard which processes the information.

One of the main features of Market Lens is its searching and sorting functionality. The client can basically search for a particular trend and sort the results based on parameters such as category, name, popularity and platform. The Spring Boot backend comes in the picture to process these requests efficiently and returns the outcome to the frontend.

We also plan to connect Market Lens with social media platforms using APIs. This will allow us to get information directly from these platforms instead of using only the data we have already saved. It will help us show new trends and updates on the dashboard.

Updated Project Approach and Architecture (2 pts)

The dashboard will collect information from social media platforms using APIs and other available sources.

A) Data Collection:
The data includes categories, popularity, likes, comments, hashtags, views, and timestamps.

B) Data Storage – MySQL:
The data is stored using MySQL, a relational database management system. It helps in storing and organizing data in chronological order for Market Lens operations.

C) Backend – Spring Boot:
The Spring Boot framework is a Java-based framework that serves as the central processing layer. It is responsible for handling core business logic, REST APIs, database communication, searching, sorting, filtering, and API requests from React.

D) Intelligent Search Using Hashing:
For quick and efficient searching, HashMap-based data structures are used to provide fast access to frequently requested information.

E) REST API Layer:
Spring Boot exposes REST endpoints that connect the frontend with backend functionality.

F) Interactive Frontend – React:
The frontend is designed using React, which provides an interactive, simple, and user-friendly interface. It allows users to explore and visualize information easily through the dashboard.
It includes features such as trend cards, hover effects, and other interactive elements.

Tasks Completed (7 pts)

Task Completed	Team Member
Designed the Market Lens dashboard using React and created different pages with simple and easy navigation.	Angel
Set up the MySQL database to store competitor details and other important information.	Suhani
Created basic REST APIs to connect the frontend and backend and manage competitor data.	Urvi
Connected the MySQL database with Spring Boot to store and access data.	Suhani
Set up the Spring Boot backend and connected it with the React frontend.	Angel
Added a feature to get competitor details from the database and show them on the Competitors page.	Urvi
Created the competitor table in the database and added some records to check if the data was saved properly.	Suhani
Added features to add, view and delete competitors directly from the dashboard.	Angel
Tested the basic features and checked if competitor details were saved and still appeared after refreshing the page.	Urvi

Challenges/Roadblocks (7 pts)

1. When we started working with MySQL, we had some trouble connecting it to Spring Boot and creating our first database table.
2. At first, we faced errors while saving competitor details in the database and getting them back.
3. We had to spend some time understanding how the Controller, Repository and Entity classes work together in Spring Boot.
4. Connecting our React dashboard with the backend was also new for us. We had to learn how REST APIs work before connecting both parts.
5. We faced a CORS error when React tried to send requests to Spring Boot. We checked the backend settings and fixed the issue.
6. Sometimes the information shown on the dashboard did not match what was saved in MySQL. We had to check the data flow and test the connection again.
7. We are still learning how to use DSA for searching and sorting. Our next challenge is to add these features to Market Lens.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Implement backend features to retrieve relevant trends and social media posts.	Angel
Learn and use Docker to run the Market Lens application on different systems.	Suhani
Implement WebSockets to provide real-time updates on the dashboard.	Urvi
Add sorting functions to the Sort By dropdown on the Market Lens dashboard.	Angel
Connect real social media data or a publicly available dataset to the database for trend and engagement analysis.	Suhani
Improve backend performance and prepare the project for deployment and presentation.	All Members
Implement HashMap-based indexing to make data searching and retrieval faster.	Urvi
Perform complete testing and fix errors in the frontend, backend, APIs and database.	All Members

Project Outcome/Deliverables (2 pts)

Market Lens is a web project that we are making to help users find social media trends in one place. We have completed some basic parts of our project.

- We made the dashboard using React.
- We created pages for trends, competitors and campaigns.
- We used Spring Boot for the backend.
- We connected React with Spring Boot using REST APIs.
- We used MySQL to save competitor details.
- Users can add, view and delete competitors.

Right now, the dashboard and competitor features are working. We still need to add real social media data, searching and sorting features, and other parts of the project.

Progress Overview (2 pts)

In the Market Lens project, 90% of the frontend is completed, still figuring out and understanding on how to improve and make it more interactive.

Backend using SpringBoot is partially completed where the data is flowing throughly from React to the springboot and 50% of the Database is integrated using MySQL. Frontend and backend integration is completed with 60% of the work.

DSA implementation, testing, debugging backend and final deployment with minimal progress on these issues so far, currently prioritising the setup. Action on these issues like Containerisation using Docker and Real time updates using web sockets is still pending.

Codebase Information (2 pts)

Repository:

https://github.com/T-053/MarketLens/tree/main

Branches:

We have not made separate branches yet. We want to use GitHub to share our work and see the changes made by each team member.

Important Work Done:

- **Angel:** Made the React dashboard, started the Spring Boot backend and added options to add, view and delete competitors.
- **Suhani:** Made the MySQL database, created the competitor table and connected the database with Spring Boot.
- **Urvi:** Worked on the REST APIs, displayed competitor details on the dashboard and tested the basic features.

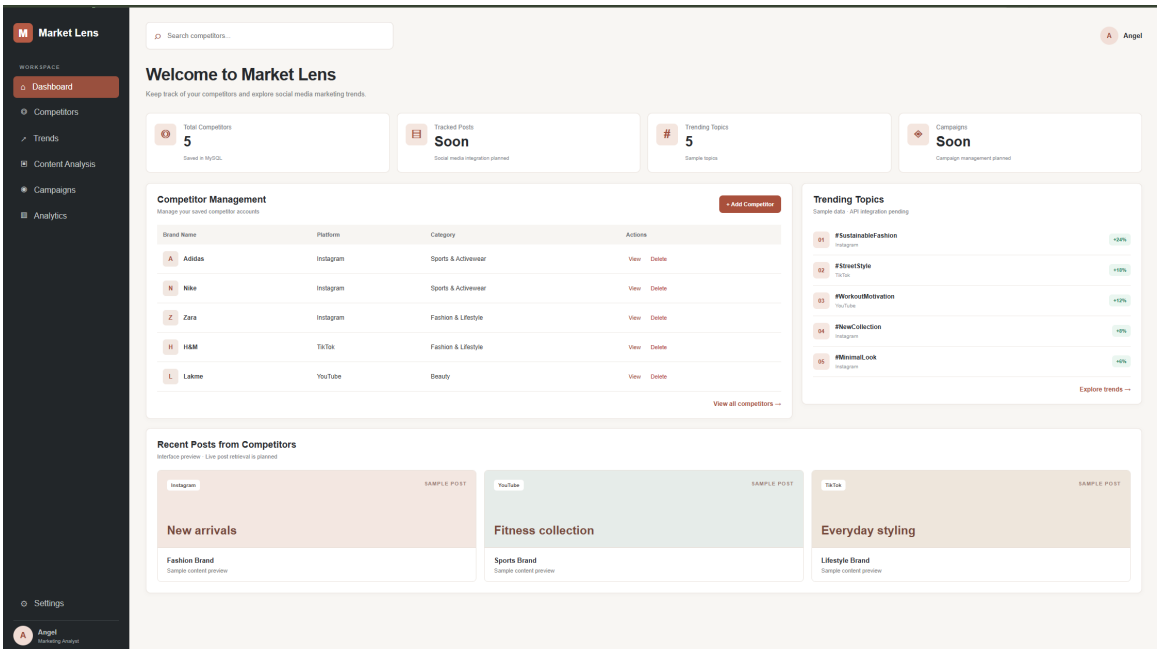
Testing and Validation Status (2 pts)

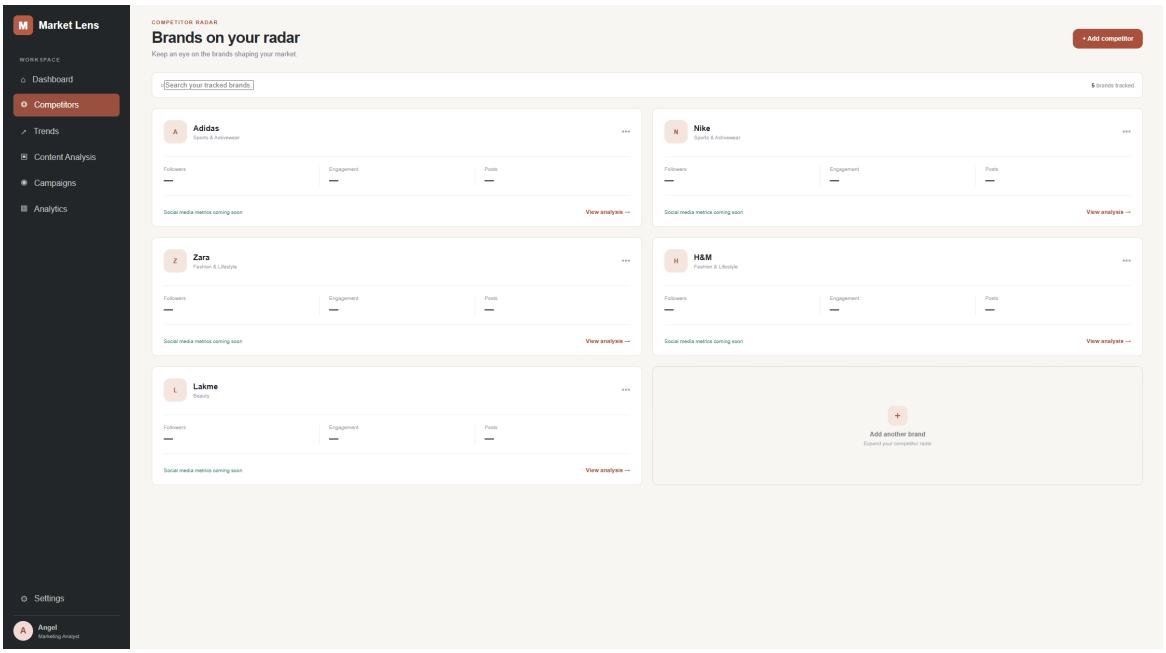
Test Type	Status	Notes
Frontend Testing	Pass	Dashboard pages are opening properly
Backend Testing	Pass	Basic Spring Boot APIs are working
Database Testing	Pass	Competitor data is saved in MySQL
Add Competitor	Pass	New competitors can be added
Delete Competitor	Pass	Competitors can be removed
Data Display	Pass	Saved competitor details appear on the dashboard
Refresh Test	Pass	Data remains saved after refreshing the page
Complete Testing	Pending	Remaining features still need to be tested

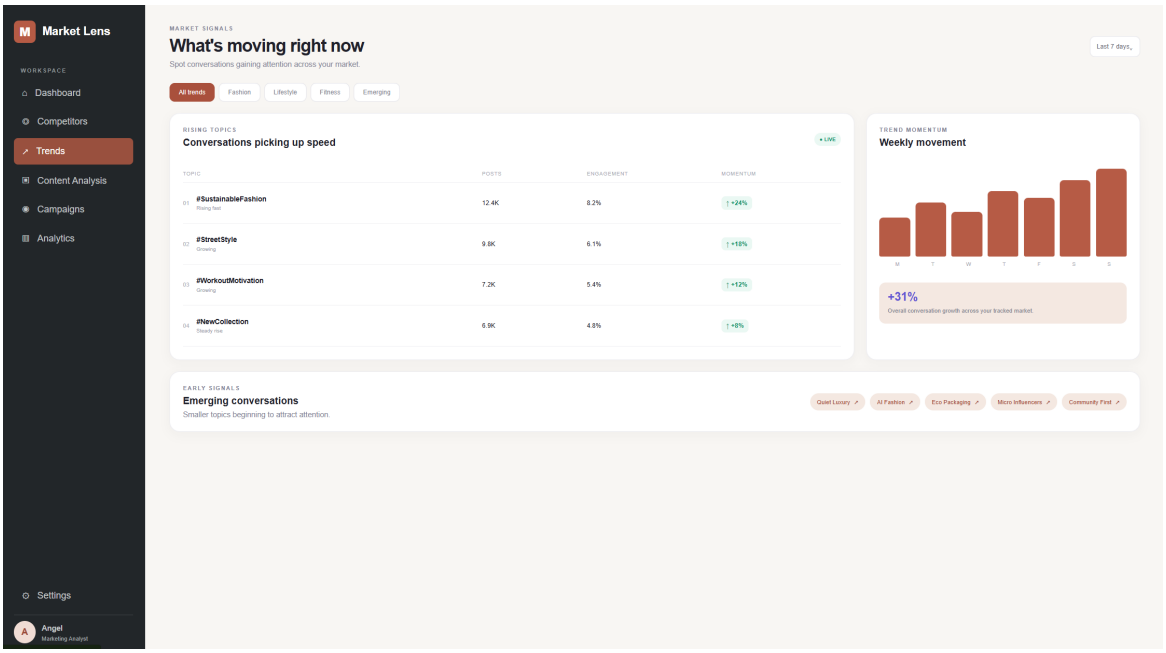
Deliverables Progress (2 pts)

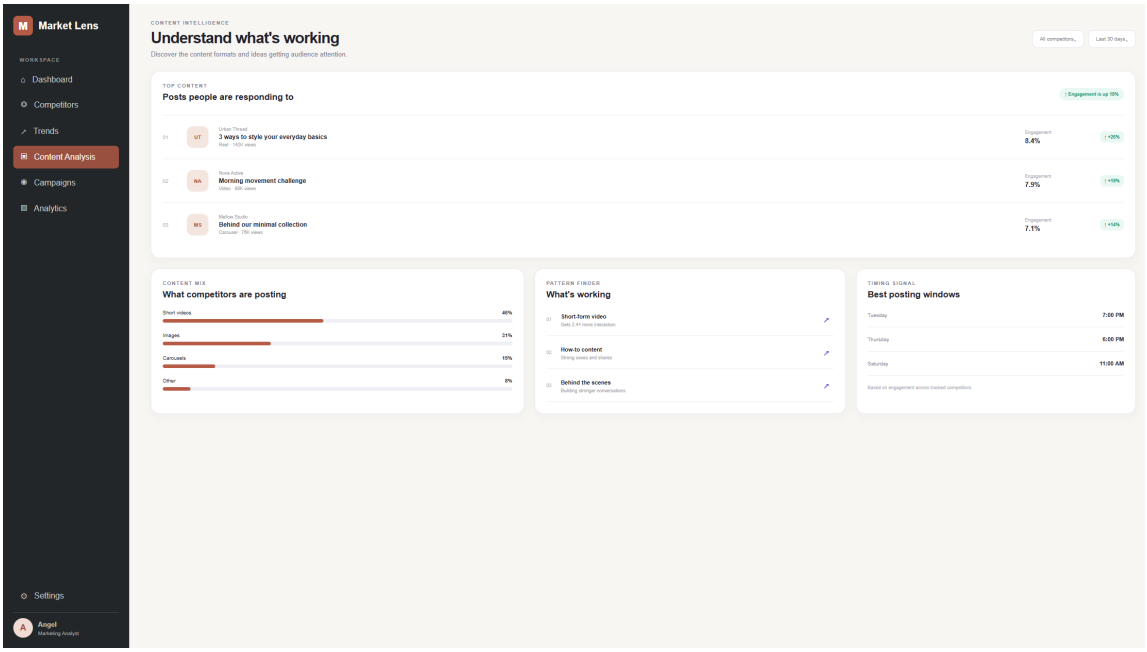
- React dashboard and different pages: **Completed**
- Spring Boot backend and basic REST APIs: **Completed**
- MySQL database and competitor management: **Completed**
- Connecting remaining features with the backend: **In Progress**
- DSA based searching, sorting and social media data: **Pending**
- WebSockets, Docker and final deployment: **Pending**

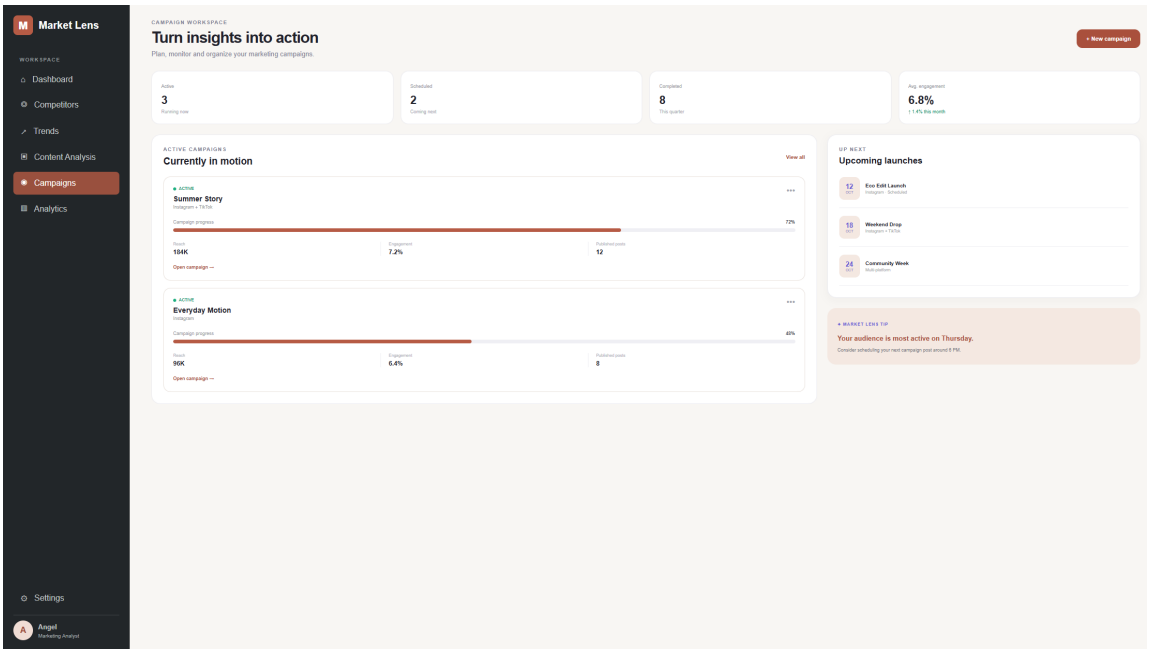
Project Screenshots

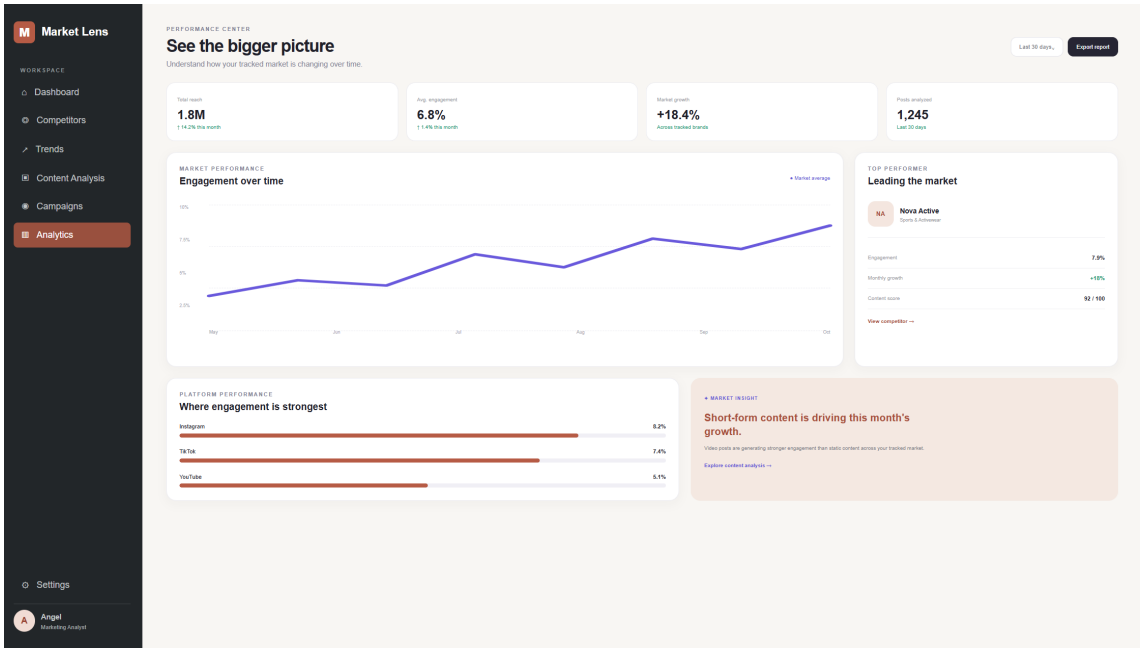





Market Lens
WORKSPACE
Dashboard
Competitors
Trends
Content Analysis
Campaigns
Analytics
Settings
Angel
Marketing Analyst
MARKET SIGNALS
What's moving right now
Spot conversations gaining attention across your market.
Last 7 days
All trends
Fashion
Lifestyle
Fitness
Emerging
RISING TOPICS
Conversations picking up speed
• LIVE
TOPIC
POSTS
ENGAGEMENT
MOMENTUM
01 #SustainableFashion
Rising fast
12.4K
8.2%
↑ +24%
02 #StreetStyle
Growing
9.8K
6.1%
↑ +18%
03 #WorkoutMotivation
Growing
7.2K
5.4%
↑ +12%
04 #NewCollection
Steady rise
6.9K
4.8%
↑ +8%
TREND MOMENTUM
Weekly movement
M T W T F S S
+31%
Overall conversation growth across your tracked market.
EARLY SIGNALS
Emerging conversations
Smaller topics beginning to attract attention.
Quiet Luxury ↗
AI Fashion ↗
Eco Packaging ↗
Micro Influencers ↗
Community First ↗


Market Lens
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Dashboard
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Marketing Analyst
CONTENT INTELLIGENCE
Understand what's working
Discover the content formats and ideas getting audience attention.
All competitors
Last 30 days
TOP CONTENT
Posts people are responding to
↑ Engagement is up 18%
01 UT
Urban Thread
3 ways to style your everyday basics
Engagement
8.4%
↑ +20%
02 NA
Nova Active
Morning movement challenge
Engagement
7.9%
↑ +19%
03 MS
Mellow Studio
Behind our minimal collection
Engagement
7.1%
↑ +14%
CONTENT MIX
What competitors are posting
Short videos 46%
Images 31%
Carousels 15%
Other 8%
PATTERN FINDER
What's working
01 Short-form video
Gets 2.4× more interaction
02 How-to content
Strong saves and shares
03 Behind the scenes
Building stronger conversations
TIMING SIGNAL
Best posting windows
Tuesday 7:00 PM
Thursday 6:00 PM
Saturday 11:00 AM
Based on engagement across tracked competitors


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WORKSPACE
Dashboard
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Marketing Analyst
CAMPAIGN WORKSPACE
Turn insights into action
Plan, monitor and organize your marketing campaigns.
+ New campaign
Active
3
Running now
Scheduled
2
Coming next
Completed
8
This quarter
Avg. engagement
6.8%
↑ 1.4% this month
ACTIVE CAMPAIGNS
Currently in motion
View all
ACTIVE
Summer Story
Instagram + TikTok
Campaign progress
72%
Reach
184K
Engagement
7.2%
Published posts
12
Open campaign →
ACTIVE
Everyday Motion
Instagram
Campaign progress
48%
Reach
96K
Engagement
6.4%
Published posts
8
Open campaign →
UP NEXT
Upcoming launches
12 OCT
Eco Edit Launch
Instagram · Scheduled
18 OCT
Weekend Drop
Instagram + TikTok
24 OCT
Community Week
Multi-platform
+ MARKET LENS TIP
Your audience is most active on Thursday.
Consider scheduling your next campaign post around 6 PM.


Market Lens
WORKSPACE
Dashboard
Competitors
Trends
Content Analysis
Campaigns
Analytics
Settings
Angel
Marketing Analyst
PERFORMANCE CENTER
See the bigger picture
Understand how your tracked market is changing over time.
Last 30 days
Export report
Total reach
1.8M
↑ 14.2% this month
Avg. engagement
6.8%
↑ 1.4% this month
Market growth
+18.4%
Across tracked brands
Posts analyzed
1,245
Last 30 days
MARKET PERFORMANCE
Engagement over time
+ Market average
16%
7.5%
5%
2.5%
May
Jun
Jul
Aug
Sep
Oct
TOP PERFORMER
Leading the market
NA
Nova Active
Sports & Activewear
Engagement
7.9%
Monthly growth
+18%
Content score
92 / 100
View competitor →
PLATFORM PERFORMANCE
Where engagement is strongest
Instagram
8.2%
TikTok
7.4%
YouTube
5.1%
+ MARKET INSIGHT
Short-form content is driving this month's growth.
Video posts are generating stronger engagement than static content across your tracked market.
Explore content analysis →